From Acceptance Specs to Unit Specs

We should take a moment to revisit what we mean by "unit specs." The common term *unit testing* means different things to different people—or even to the same person on different projects. Unit tests typically involve isolating a class or method from the rest of the code. The result is faster tests and easier-to-find errors.

Where unit testing approaches differ is the *degree* of isolation; that is, whether to remove every possible dependency or to test a related group of collaborating objects together. In this book, we'll use *unit spec* to refer to the fastest, most isolated set of tests for a particular project. For more information, see Martin Fowler's article on the subject.[1]

With the unit tests in this chapter, you won't be calling methods on the `API` class directly. Instead, you'll still be simulating HTTP requests through the `Rack::Test` interface. Typically, you test a class through its public interface, and this one is no exception. The HTTP interface *is* the public interface.

Drive Each Layer's Public API



Your tests for any particular layer—from customer-facing code down to low-level model classes—should drive that layer's public API. You'll find yourself making more careful decisions about what does or doesn't go into the API. Moreover, your tests will be less brittle and will give you greater latitude for refactoring your code.

You will, however, be isolating the public-facing API from the underlying storage engine. In effect, you're testing one layer of the app at a time. In this chapter, that means driving the behavior of the `API` class that routes incoming requests to the storage engine. In the next chapter, you'll test and design the storage engine itself.

For more on the nuances among different kinds of specs, see Xavier Shay's article, "How I Test Rails Applications."[2]

A Better Testing Experience

Before jumping into the tests, let's take a moment to configure RSpec for the task at hand. The default RSpec setup is minimal by design. But the framework offers a number of suggested settings that are easy to turn on.

1. https://martinfowler.com/bliki/UnitTest.html
2. https://rhnh.net/2012/12/20/how-i-test-rails-applications/